

PUJA MANKAR

puja.mankar.k@gmail.com | (+91)8208731193 | [linkedIn/puja-mankar-819b1b17b/](https://www.linkedin.com/in/puja-mankar-819b1b17b/)

PROFILE SUMMARY

Microbiologist with R&D and regulatory experience in food formulation, scale up, analytical testing, and compliance. Skilled in assay validation, process optimization, and ensuring nutritional labeling accuracy aligned with FSSAI and global standards.

WORK EXPERIENCE

VALCORE FOODS PVT LTD | RESEARCH AND DEVELOPMENT MANAGER

| July 2025 – Oct 2025

- Led fermentation studies on chilli mash, optimizing LAB dominance under varying methodologies, salt concentrations, particle sizes, and anaerobic conditions.
- Designed preservative efficacy trials (benzoate, sorbate, metabisulphite, ascorbic acid, tocopherols, EDTA, blanching) to evaluate microbial stability, color retention, and shelf life.
- Conducted and Standardized analytical assays
- Developed troubleshooting protocols for fermentation failures (kahm yeast, texture loss, off-flavors) and generated reproducible datasets for process refinement.
- Working on developing a fermented hot sauce with focus on stability, sensory profile, texture, packaging, and physicochemical characteristics in alignment with FSSAI guidelines.
- Investigated pigment degradation in green and red peppers linked to processing parameters and metal ion content.

ANUVI FOOD SCIENCES PVT LTD | BUSINESS OPERATIONS EXECUTIVE

| Feb 2024 – June 2025

- Scaled in-house production of fermentation-based food additives (6x and 10x) by optimizing experimental design.
- Reviewed and updated patents to protect product design and intellectual property (IP).
- Served as the primary contact for regulatory compliance and license acquisition, including FSSAI and HALAL certifications.
- Designed POC to evaluate the efficacy of bitter blocker (a taste modulator) in enhancing the palatability of stevia. Following validation, expanded the design to accommodate use cases like chocolate coffee and tea.
- Developed and managed a commercially scalable production methodology ensuring viability and large-scale implementation, for the same.
- Designed and analyzed sensory evaluation experiments for cell-derived additives.

MYLAB DISCOVERY SOLUTIONS | INTERN

| Feb 2023 - Aug 2023

- Collaborated with BD Max to optimize in-house RT-PCR kits for end-to-end automation.
- Automated RT-PCR kits on in-house machines, including Compact XL, Compact Dx, and Compact Q.
- Worked on the development and optimization of an in-house RT-PCR kit for HHH detection.
- Part of the validation team for RT-PCR kits, ensuring accuracy and regulatory compliance.

TECHNICAL SKILLS

- Microbial culture isolation, maintenance, and sampling
- Experiment design and protocol development (SOPs, safety standards)
- Analytical assays: titration, spectrophotometry, RT-PCR, microbial profiling
- Quality control, documentation, and regulatory compliance (GLP, FSSAI, ISO)
- Team training, equipment maintenance, and lab workflow management

PROJECTS

NGS-SOIL METAGENOMICS | RESEARCH ASSISTANT

| Oct 2021- Sept 2022

Analyzed NGS data to characterize and identify soil metagenomes across different cropping systems and fertilizers.

COLLEGE VICINITY FOOD SAFETY ASSESSMENT | STUDENT

| Aug 2017- Dec 2017

Isolated, identified, and characterized bacteria responsible for food poisoning at outlets frequented by college(Fergusson College, Pune) students. Successfully detected Staphylococcus aureus, Salmonella, and Campylobacter.

EDUCATION

MASTERS IN MICROBIOLOGY

CGPA 9.31 | July 2021

MODERN COLLEGE OF ARTS, SCIENCE AND COMMERCE, PUNE

BACHELORS IN MICROBIOLOGY

CGPA 7.35 | July 2019

FERGUSSON COLLEGE, PUNE

ACHIEVEMENTS & EXTRA-CURRICULARS

- Certification on Food Safety Training (FoSTaC)
- Certification on ISO 22000:2018 FSMS Lead Auditor
- GLP/EHS Laboratory Practices (Training, 2023)
- Certification course on techniques in Pharmaceutical Microbiology.
- Certification course on working of Pharmaceutical industry, Quality assurance and Quality control.
- Volunteering for Covid Test initiative at Molecular Diagnostic Research Centre of Savitribai Phule University.